

Digit Recognition using HOG-SVM & PCA

Project Goal: High-Accuracy Handwritten Digit Classification

Team Members:

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The Challenge: Why Raw Pixels Fail

- **Dataset:** MNIST (Modified National Institute of Standards and Technology) , 70,000 grayscale images, size 28x28 pixels.
- Raw pixels are sensitive to noise.
- Small variations (shifts, rotations, line thickness) drastically change raw pixel values.
- **Requirement:** Need invariant features that capture the digit's structure, not just its location.

Feature Engineering with HOG

- **Tool:** Histogram of Oriented Gradients (HOG)
- **Principle:** Captures local shape by analyzing edge direction and intensity.
- **Why Gradients?** Gradients effectively highlight contours and edges, defining the digit's skeleton.
- **Parameters:** 9 orientations, 7x7 pixels per cell, 2x2 cells per block.

Dimensionality Reduction via PCA

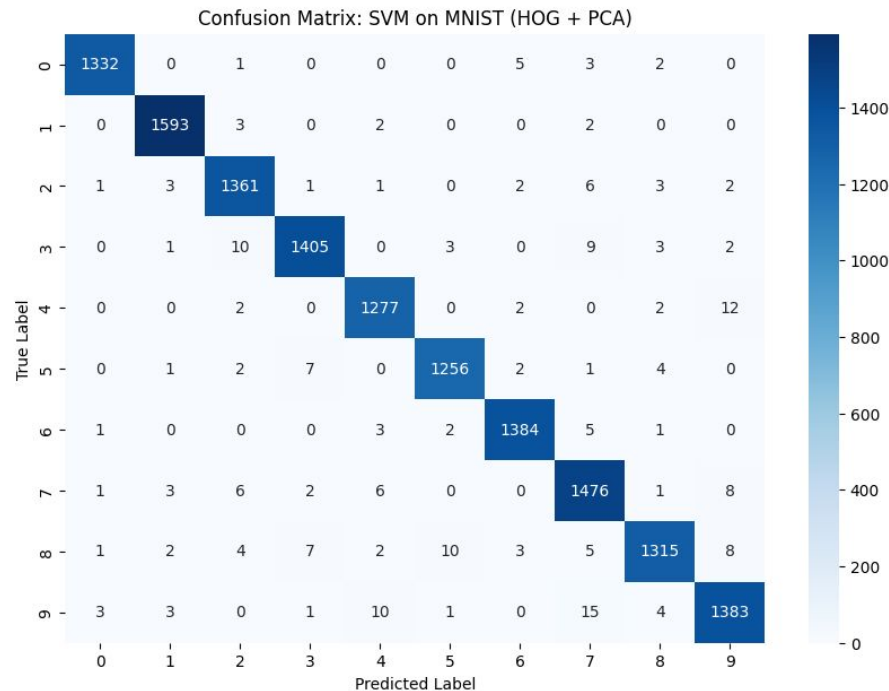
- **Problem:** HOG output is still a high-dimensional feature vector, slowing down training.
- **Solution:** Principal Component Analysis (PCA) is applied to the HOG features.
- **Necessity:** Reduces feature space to increase speed and mitigate overfitting from noise.
- **Method:** Finds the orthogonal components (eigenvectors) that maximize data variance.
- **Result:** Features reduced to 100 components, retaining 95% of the original variance.

Modeling with SVM and RBF Kernel

- **Classifier:** Support Vector Machine (SVM)
- **Kernel Selection:** Radial Basis Function (RBF) Kernel.
- **Why RBF?** MNIST features are non-linearly separable; RBF maps data to a higher dimension to find the optimal separating hyperplane.
- **Hyperparameters:** C=10 (regularization), Gamma='scale'.
- **Model Persistence:** Trained models and PCA transformer serialized into `.pkl` files.

Performance and Evaluation Metrics

- Model achieved a 98% accuracy on the test set.
- **Validation Tool:** Confusion Matrix is essential for deep performance analysis.
- Confusion Matrix reveals which specific digits are often confused by the model.
- Confirms high precision and recall, minimizing common errors like confusing 4s and 9s.



Deployment with Streamlit

- **Tool:** Streamlit web framework used to develop the user interface.
- **UI Feature:** Interactive drawing canvas for real-time user input.
- When a user draws, the app instantly runs the image through the HOG-PCA-SVM pipeline to show a prediction.
- **Function:** Provides immediate, practical demonstration of the trained model.

The model correctly predicting a number



Hand-Drawn Digit Classifier

Powered by HOG Features + SVM

Draw a digit (0-9) in the center of the box below.



Classify Digit

The model thinks this is a: 5



Hand-Drawn Digit Classifier

Powered by HOG Features + SVM

Draw a digit (0-9) in the center of the box below.



Classify Digit

The model thinks this is a: 7

Conclusion: Reviewing the ML Lifecycle

- **Exploration:** Data acquisition (MNIST) and initial analysis .
- **Feature Engineering:** Structural representation via HOG .
- **Modeling & Optimization:** Dimensionality reduction (PCA) and training (RBF-SVM) .
- **Deployment:** Interactive web application (Streamlit).
- **Process Management:** Robust GitHub infrastructure and peer review .

THANK YOU